Sumin Hong

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Summary

I build and evaluate machine learning systems on real-world data, from fine-tuned language models for retrieval to reproducible neuroimaging and microscopy pipelines, with a focus on models that generalize to unseen data and are trustworthy enough for health applications.

Education

Aug 2026 – Present — M.S. in Artificial Intelligence (Biomedical Concentration), Columbia University — New York, NY
Coursework: *Natural Language Processing, Statistical Machine Learning for Genomics, Deep Learning for Biomedical Signal Processing, Machine Learning for OR and FE*
Expected Graduation: May 2027

Jun 2026 — B.S. in Bioengineering (Data Science), University of Washington — Major GPA: 3.84/4.00
Minors: Neural Computation and Engineering; Applied Mathematics | Dean's List; Department Honors

Research Experience

Genomic Language Model Interpretability | AlQuraishi Lab, Columbia University — Sep 2026 – Present
Research Assistant, Advisor: Dr. Mohammed AlQuraishi — *New York, NY*

- **Problem:** Tested whether the lab's three genomic language models (up to 661M parameters) encode RNA secondary structure by rebuilding Minerva's RNA base-pairing benchmark (Rfam 15.0; 672 families, 6.1M candidate pairs), where default training underperformed a single untrained attention head
- **Insight:** Showed that the base-pairing signal sits in a few attention heads and that default head-weight training was undertrained; found that feature standardization, a common default, made Minerva-style regression overfit (305 of 350 heads used), since raw attention scale already marks informative heads
- **Result:** Raised the 35-layer model's test AUPRC from 0.457 to 0.486 (about 80× random) by caching per-head attention maps (each training run from hours to seconds) and searching 90–126 configurations per model; matched Minerva's released code for a fair comparison (0.435 to 0.475); ran all jobs on Columbia's Insomnia HPC cluster (SLURM, GPUs)

Simultaneous EEG-fMRI Analysis Across Wake and Sleep | LIINC, Columbia University — Sep 2026 – Present
Research Assistant, Laboratory for Intelligent Imaging and Neural Computing, Advisor: Dr. Paul Sajda — *New York, NY*

- **Problem:** EEG recorded inside the MRI scanner (32 channels, 5 kHz, 255 sessions, 33 subjects) was dominated by scanner artifacts about 20× larger than brain signal, and its sleep labels and EEG-fMRI timing had not been verified
- **Insight:** Systematic QC showed that 44% of resting-state epochs were scored as sleep, deep sleep was too sparse to model (42 of 6,990 epochs), and one subject was recorded without scanner clock sync, revealed by trigger drift that grew with session length
- **Result:** Implemented MRI gradient artifact removal (average artifact subtraction) from scratch in Python within a reproducible QC pipeline, reducing channel amplitude from ~700–1,100 µV to ~30–55 µV, and defined a balanced wake vs sleep target (3,344 vs 3,515 epochs) for modeling

Segmentation-Free ML for Endothelial Junction Phenotyping | UW Bioengineering — Aug 2023 – Sep 2026
Research Assistant, Advisor: Dr. Ying Zheng — *Seattle, WA*

- **Problem:** Junction morphology in engineered 3D brain microvessels predicts barrier leakiness but was scored by eye; standard cell segmentation failed on dense 3D confocal images, leaving drug screens across 300+ biological replicates without a quantitative readout
- **Insight:** Showed that segmentation is unnecessary because junction shape can be read from intensity profiles between neighboring nuclei, which Cellpose detects reliably; gated measurements by signal strength (MAD z-scores) after finding that naive peak tracing turned background noise into false "wavy" junctions
- **Result:** Built a Python pipeline (Cellpose, NumPy, pandas, scikit-learn) extracting 5 junction features across 182 images from 4 experiments, clustered phenotypes with Gaussian mixture models (k by silhouette score), and ranked 25 compounds on a thrombin-to-control rescue axis; co-first author on a manuscript in preparation, contributed to a bioRxiv preprint and a PNAS submission, and presented at 7 venues including ISSCR

Projects & Hackathons

Cross-Project Bug Triage with Fine-tuned Language Models | COMS 4705 (NLP), Columbia — Nov 2026 – Present
Team Lead (3 members), in progress — *New York, NY*

- **Problem:** Bug triage models are usually trained and tested on the same project, yet a deployed tool must handle unseen codebases; a pilot showed BM25 Recall@10 falling from 0.50 on Firefox to 0.19 on VS Code

- **Research Design:** Led a 3-person team; defined the cross-project question and designed a leakage-free evaluation harness with temporal splits and cross-tracker transfer tests (Bugzilla, Jira, GitHub) over 100K+ bug reports from 8 open-source projects
- **Retriever:** Building a dense duplicate-bug retriever by contrastively fine-tuning a sentence embedding model with hard negatives in a custom PyTorch training loop, benchmarked against BM25 and Sentence-BERT on Recall@10 and MRR

Trust Issues: How to Disrupt AI Cartels | **Paper** | CAIAC Multi-Agent Collusion Hackathon — Oct 2026
Technical Fellowship Member, Columbia AI Alignment Club (CAIAC); 4-person team — *New York, NY*

- **Experiment:** Isolated the driver of cartel breakdown among LLM agents with a talk-vs-action control experiment in a 3-agent market (Qwen 3.5 9B via Ollama): greedy offers alone kept prices at the cartel level (52), while silent overproduction crashed them to cost (4) within one round across 2 independent runs; featured as a key figure in the submitted paper
- **Engineering:** Added 3 experimental conditions to the team's Python simulation framework by decoupling an agent's stated offer from its actual action across the protocol, engine, and reporting modules, with exact-text delivery checks and all existing regression tests passing
- **Analysis:** Identified a retaliation threshold (agents tolerated up to 17% profit loss but retaliated at 33%) in a 4-condition parameter sweep, and flagged a say-do gap where agents pledged cooperation while producing at retaliation levels

Publications

Shin, Y.J.*, **Hong, S.***, Lee, J.D., Zheng, Y. Segmentation-Free, Pixel-Level Machine Learning Reveals Distinct Endothelial Junction Phenotypes in Traumatic Brain Injury Models. (in preparation)

Shin, Y.J., Wei, L., **Hong, S.**, et al., Zheng, Y. (2025). A Machine Learning-3D Microvessel Platform Identifies Kinase Targets Restoring Blood-Brain-Barrier Endothelial Integrity. *bioRxiv*. doi:10.64898/2025.12.11.692686

Shin, Y.J., Wei, L., **Hong, S.**, et al., Zheng, Y. Polypharmacological Kinase Screening and Prediction in 3D Brain Microvessel Models for Enhancing Barrier Integrity in Traumatic Brain Injury. *PNAS* (under review)

Shin, Y.J., Evitts, K., et al., **Hong, S.**, Hadland, B., Zheng, Y. Perfusable Vascular Organoids for Modeling Hierarchical Cerebrovasculature. (in preparation)

** Equal contribution*

Teaching

Sep 2025 – Jun 2026	**Teaching Assistant, Neural Engineering (BIOEN 460/560, CSE 490)** and **Neural Engineering Tech Studio (BIOEN 461/561),** University of Washington: graded ML programming assignments for 100+ students and mentored 30+ student teams from concept to prototype

Technical Skills

Languages	Python, R, MATLAB
Machine Learning	PyTorch, scikit-learn, contrastive learning, attention analysis, Gaussian mixture models, LLM agents (Ollama)
Data	NumPy, pandas, Matplotlib, ggplot2, statistical modeling and hypothesis testing
Signals & Imaging	EEG-fMRI preprocessing and artifact removal, Cellpose, confocal image analysis
Systems	Git/GitHub, Linux, HPC clusters (SLURM, GPUs), Jupyter

Awards & Honors

2025 – 2026	**Levinson Emerging Scholars Award**	University of Washington
Jun – Aug 2025	**Summer Undergraduate Research Fellowship (ISCRM)**	University of Washington
Mar – Aug 2025	**Mary Gates Research Scholarship**	University of Washington